

1. Show that the sliced score matching (SSM) loss can also be written as

$$L_{SSM} = \mathbb{E}_{x \sim p(x)} \mathbb{E}_{v \sim p(v)} [\|v^T S(x; \theta)\|^2 + 2v^T \nabla_x (v^T S(x; \theta))].$$

by def :

$$L_{SSM} = \mathbb{E}_{x \sim p(x)} \|S(x; \theta)\|^2 + \mathbb{E}_{x \sim p(x)} \mathbb{E}_{v \sim p(v)} (2v^T \nabla_x (v^T S(x; \theta)))$$

$$\mathbb{E}_v [\|v^T S\|^2] = \mathbb{E}_v [(v^T S)^2] = S^T \mathbb{E}_v (v v^T) S$$

$$\text{if } v \sim \mathcal{N}(0, I), \text{ then } \mathbb{E}_v [v v^T] = I$$

$$\text{so, } \mathbb{E}_v [\|v^T S\|^2] = \|S\|^2 \quad \text{--- (1)}$$

$$\text{Let } f(x) = v^T S(x; \theta), \quad \nabla_x f(x) = (\nabla_x S(x; \theta))^T v$$

$$\text{so, } v^T \nabla_x (v^T S) = v^T (\nabla_x S)^T v = v^T (\nabla_x S) v$$

$$\mathbb{E}_{x,v} [2v^T \nabla_x (v^T S)] = \mathbb{E}_{x,v} [2v^T (\nabla_x S) v] \quad \text{--- (2)}$$

by ① ②

$$L_{SSM} = \mathbb{E}_x \|S\|^2 + \mathbb{E}_{x,v} [2v^T \nabla_x (v^T S)]$$

$$= \mathbb{E}_{x,v} \|v^T S\|^2 + \mathbb{E}_{x,v} [2v^T \nabla_x (v^T S)]$$

$$= \mathbb{E}_{x,v} [\|v^T S\|^2 + 2v^T \nabla_x (v^T S)] \quad \#$$

2. Briefly explain SDE.

SDE (Stochastic Differential Equation) :

a differential equation that describes a random process evolving over time t . the modeling systems with both deterministic trend and random fluctuations. The general form :

$$dx_t = \underbrace{f(x, t)}_{\text{drift}} dt + \underbrace{G(x, t)}_{\text{diffusion}} dW_t, \quad x(0) = x_0$$

x_t : a stochastic process we solve for

In short : SDE is continuous-time engine for modeling, simulating, and generating complex stochastic processes.

use for :

(1) Model real-world randomness

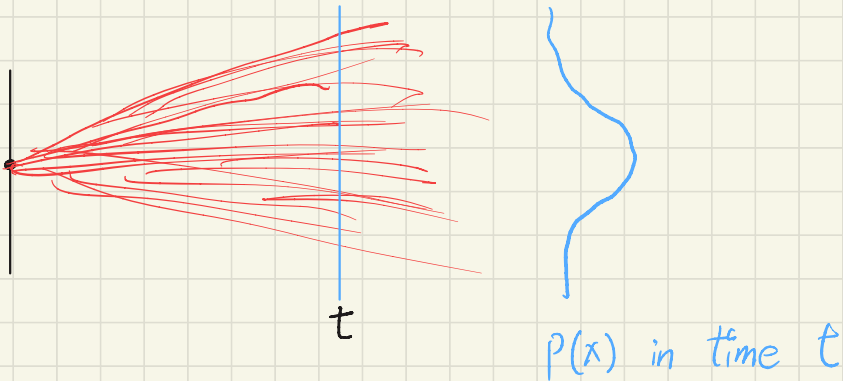
(2) Generate data from noise

(3) Simulate and predict.

3. Unanswered Questions

There are unanswered questions from the lecture, and there are likely more questions we haven't covered.

上課有一個圖



有辦法用 $p(x)$ 反推出發點嗎？

若有多個出發點如何反推？